

Ben Bubnick

Master Career History

Email: ben.bubnick@gmail.com | Phone: (216) 556-1612
LinkedIn: linkedin.com/in/benbubnick | GitHub: github.com/bubnickbf

Career Overview

Comprehensive working record spanning healthcare data consulting, data engineering, analytics, software development, scientific computing, entrepreneurship, higher education, academic research, farm operations, and community service. Combines hands-on delivery with technical leadership, documentation, teaching, stakeholder engagement, business administration, and quality-focused practices. Maintained as a source for targeted resumes, biographies, applications, and future career pivots.

Skills and Domains

Data Engineering and Software: SQL, Python, Ruby, Snowflake, dbt, SQLMesh, Spark, Hadoop/HDFS, Hive, Impala, Kafka, PostgreSQL, Oracle, C#, VB.NET, C++, FORTRAN, ETL, ELT, data warehouses, data lakes

Cloud and Delivery: AWS, Azure, Docker, Git/GitHub, GitHub Actions, Jenkins, Azure DevOps, Jira, Confluence, CI/CD, technical documentation, mentoring

Healthcare and Analytics: Clinical, payer, eligibility, revenue-cycle, and financial data; HL7, FHIR, EDI/837, HEDIS, EHR/EMR, HIPAA, HITRUST, pandas, Jupyter, Power BI, Tableau, statistical modeling

Science and Education: Spectroscopy, photometry, optical measurement, radiative-transfer modeling, Fourier analysis, Monte Carlo methods, numerical integration and interpolation, physics and astronomy instruction, curriculum development, research publication

Business Operations: Bookkeeping, accounting, tax compliance, marketing, promotional-material development, client acquisition

Earlier Scientific and Creative Tools: R, PERL, IDL, IRAF, Mathematica, MATLAB, LabVIEW, Maple, PDL, JMP, SAS/GRAPH, AutoCAD, Adobe Illustrator, Adobe Photoshop, GIMP, LaTeX

Professional Experience

HHAAExchange

Oct. 2025 - Jun. 2026

Healthcare Data Integration Consultant - Contract via Veersa Technologies

Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox

Led healthcare source analysis, data-quality investigation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains.
- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.

Adonis

Mar. 2025 - Jul. 2025

Healthcare Revenue Cycle Data Consultant - Contract via Health Data Movers

Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding, analytics, and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models supporting revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Diagnosed undocumented source-key versioning and implemented composite business grains with SHA-256 surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
- Authored implementation playbooks, technical handoff documentation, and a QA runbook for a repeatable healthcare-data validation workflow.

GrowData Analytics

Sep. 2024 - Oct. 2025

Independent Data Automation and Analytics Consultant

Technologies: Ruby, RSpec, OCR, LLM APIs, audit logging, automated testing

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable

- transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.
- Managed the business's bookkeeping, accounting, and tax obligations.
- Led marketing and client acquisition, including creating promotional materials for the business.

Arkos Health

Jun. 2023 - Jan. 2025

Senior Data Analyst

Technologies: AWS S3, Athena, Redshift, Glue, dbt, PySpark, Docker, Jupyter, Power BI, Tableau

- Architected a greenfield AWS data lake integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets.
- Integrated Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting.
- Instrumented scheduled pipelines with monitoring, alerting, and more than 10 failure classifications; authored three AWS/dbt incident runbooks supporting rapid detection and recovery.

Merative

Jul. 2022 - Mar. 2023

Senior Data Scientist

Technologies: Azure Data Factory, Blob Storage, Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps

- Engineered a seven-layer Python and SQL migration-acceptance suite covering partition reconciliation, schema compatibility, null handling, duplicates, field-level comparisons, aggregates, and business rules, achieving exact HDFS-to-Azure fidelity.
- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.

IBM

Jun. 2015 - Jul. 2022

Advanced through five roles delivering healthcare analytics, data architecture, client integration, data-quality improvement, and large-scale migration validation.

Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, PostgreSQL, Docker, Jenkins, Tableau, HL7

- Supported employee onboarding from 2016 to 2018, helping new colleagues learn team practices, systems, and technical workflows.
- Served on tiger teams supporting the HDFS-to-Azure data-platform upgrade and participated on the data-governance team.
- Participated in internal coding jams, hackathons, and cross-team delivery initiatives.
- Played a central role in the REACH cross-team training initiative, teaching SQL fundamentals to nontechnical colleagues and ETL, architecture, data-science, and predictive-modeling practices to technical colleagues.

Senior Data Scientist

Nov. 2021 - Jul. 2022

- Developed a surgical-scheduling optimizer and CNN/RNN case-card prediction proof of concept, automating case-card generation and resource allocation across operating rooms, time slots, and personnel.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.

Senior Data Architect

Dec. 2020 - Nov. 2021

- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.
- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.

Data Architect

Nov. 2017 - Dec. 2020

- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.
- Led a multi-organization healthcare integration spanning 70+ EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules and coordinating delivery with engineering teams.

Software Developer, Data Platform

Sep. 2016 - Nov. 2017

- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating \$30,000 in annual vendor costs.
- Built a Python, Ruby, and SQL extraction tool that reduced total client-integration effort by 28%, measured through eliminated story points and lower average engineering hours per implementation.

Data Scientist

Nov. 2015 - Sep. 2016

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.

Explorys, an IBM Company

Jun. 2015 - Nov. 2015

Data Science Consultant - Contract via Randstad USA*Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins*

Supported clinical-data integration and population analytics across more than 10 client data warehouses. IBM acquired Explorys in 2015; the overlap with the IBM tenure reflects the acquisition and transition rather than two unrelated concurrent roles.

- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.

AmTrust Financial Services

Mar. 2014 - Jun. 2015

Application Development Consultant - Contract via TEKsystems*Technologies: VB.NET, ASP.NET, C#, Web APIs, CSS, JavaScript, HTML5, SQL, PL/SQL*

Developed ISO-based web forms, multi-tier MVC components, Web APIs, and user interfaces for commercial insurance policy software.

- Led development work for insurance policy software, a new product offering, and the business integration of acquisitions.
- Analyzed more than 65 rating forms to identify and resolve performance bottlenecks.
- Introduced C# unit testing for multi-tier MVC components and helped train new staff while coordinating daily work allocation.

PPG Industries

Apr. 2012 - Aug. 2013

Color Scientist*Technologies: VB.NET, C#, SQL, C++, FORTRAN, JMP, spectrophotometry, numerical methods*

Developed software and radiative-transfer models for automotive paint formulation, spectroscopic profile analysis, and technology-risk assessment.

- Improved a color-matching algorithm by 40% using numerical integration methods.
- Built cross-instrument correlation software in VB.NET and JMP to improve measurement consistency and support statistical risk analysis.
- Debugged a VB.NET and FORTRAN optimization routine for a numerical radiative-transfer solution.
- Wrote a Visual C++ and VB.NET DLL wrapper for a loaned spectroscopic instrument and developed formulation methods for an automated color-matching facility.

Teaching and Academic Research**Lorain County Community College**

Aug. 2016 - May 2019

Adjunct Professor - Taught physics laboratory and astronomy lecture courses.

Miami University

Aug. 2010 - Jun. 2011

Adjunct Professor - Developed and taught Physics and Society, introductory physics, and Astronomy and Space Physics courses for more than 100 students.

Cincinnati State Community College

Feb. 2011 - Jun. 2011

Adjunct Professor - Taught technical physics for remedial students; wrote laboratory instructions and coordinated measurement, analysis, and technical-report work for five student teams.

University of Cincinnati

Oct. 2008 - Jun. 2010

Graduate Teaching Assistant - Administered approximately 60 algebra- and calculus-based physics laboratories for more than 100 engineering and non-physics students; taught measurement, Microsoft Excel, and DataStudio.

University of Cincinnati

Oct. 2009 - Dec. 2009

Engineering Teaching Assistant - Prepared and delivered Mathematica lectures for advanced aerospace-engineering students.

University of Cincinnati

Jun. 2006 - Dec. 2006

Supplemental Instruction Leader - Led peer-learning sessions of up to 12 students in calculus- and algebra-based introductory physics.

University of Cincinnati

Jan. 2006 - May 2006

Undergraduate Teaching Assistant - Supported group recitations in freshman engineering physics through demonstrations, short lectures, and individual assistance.

University of Cincinnati Learning Communities Program

Jun. 2006 - Aug. 2008

Peer Leader - Connected first-year students with faculty, academic advisers, program staff, and campus resources; mentored their transition to university life and introduced academic tools, time-management, study, and community-building skills.

Private Tutoring

Jun. 2006 - Aug. 2008

Private Tutor - Tutored honors and introductory physics students in course concepts and problem-solving techniques. Across these institutions, used interactive, peer-led, and just-in-time teaching methods; produced more than 700 hours of recorded lectures and laboratory material; and exceeded evaluation benchmarks by 16–17% year over year.

University of Cincinnati

Sep. 2006 - Sep. 2010

Research Assistant, Department of Physics - Conducted near-infrared astronomical research using observations from the 0.8–5.4 micrometer SpeX spectrograph at the NASA Infrared Telescope Facility.

- Used photometric data to estimate the distance, extinction, and physical properties of open star clusters and help map the structure of the Milky Way.
- Located more than 100 stars in each observed field and calculated radial velocities using physical and Fourier cross-correlation methods.
- Performed spectroscopy, photometry, classification, data reduction, signal analysis, and noise reduction using IRAF, IDL, Mathematica, PERL, and related scientific tools.
- Volunteered 120 hours of quality control for the Cambridge Astronomical Survey Unit Variable Stars database.
- Operated the SpeX spectrograph as a visiting astronomer at the NASA Infrared Telescope Facility on Mauna Kea in November 2007.
- Published research on near-infrared astrophysics and received the Mary J. Hanna Research Fellowship in 2010.

Ohio University

Oct. 2003 - May 2004

Undergraduate Student Researcher, Printmaking - Used a competitive undergraduate research grant to develop a lower-toxicity photo-etch process that replaced conventional photochemical materials with toner and a wintergreen-based solvent.

Earlier Employment

Lincoln Park Pub

Aug. 2013 - Mar. 2014

Cook - Worked in the restaurant kitchen.

University of Cincinnati Geology Math Physics Library

Mar. 2006 - Sep. 2008

Student Librarian - Assisted patrons with research and reference needs, maintained Library of Congress-classified materials, conducted inventory, and located missing items.

FranklinCovey

Jun. 2005 - Apr. 2006

Productivity Consultant and Sales Associate - Advised customers on planning products and workshops, used detailed product knowledge to meet client needs, and supported cash and credit transactions.

Retail Food Service and Manual Work

1995 - 2010

Various roles - Worked as a Target cashier; Dillard's sales associate; Aladdin's Eatery dishwasher; Cravings Cafe and Take 5 Coffee Company barista; Pickle Bill's Lobster House cook; and Pier W dishwasher and cook. Other early work included retail sales, coffee-shop management, telemarketing, industrial labor, FedEx loading, interior painting, cement work, yard work, swim instruction, and grocery and restaurant service.

The customer-facing and operations roles included opening and closing, training coworkers, ordering and receiving inventory, stocking and heavy lifting, handling cash and credit transactions, preparing food and espresso drinks, and resolving customer concerns during high-volume service.

Farm and Hands-On Work

Doodlebug Farms

Sep. 2024 - Present

Family Farm Operator - Runs a family farm with 80 trees and manages its ongoing expansion.

Volunteer and Community Experience

HSEA

Feb. 2025 - May 2025

ETL Brigade Volunteer - Built a reusable PostgreSQL model for community-health analytics covering claims, visits, and eligibility.

CSforAll

Oct. 2018 - May 2019

Instructor - Supported computer-science instruction for high-school students, with emphasis on learners without reliable home computer or Internet access.

HIT in the CLE

Mar. 2017 - Mar. 2019

Coach - Used a data-science competition to introduce students to health-information-technology careers and support the regional talent pipeline.

Code for America

Jan. 2015 - Jun. 2015

ETL Guild Volunteer - Applied data-science tools to improve and expand public-record access.

Heights Observer, Cleveland Heights, Ohio

Feb. 2012 - May 2014

Volunteer Editor - Edited community-newspaper content in support of local civic engagement and an independent free press.

The News Record, University of Cincinnati

Feb. 2008 - Sep. 2010

Editorial Contributor - Wrote and edited content for the university's student newspaper.

TCP World Academy

Sep. 2008 - May 2009

Tutor - Helped students connect classroom learning with participatory society-simulation activities.

Project Astro at Ethel M Taylor Academy

2008

Volunteer Astronomer - Supported a national program connecting astronomers with educators to strengthen astronomy and physical-science instruction.

Reach Out on Campus at Ohio University

Sep. 2000 - Jun. 2004

Volunteer - Participated in campus-based community service.

Leadership and Professional Affiliations

Detroit Color Council Member	2012 - 2013
American Institute of Aeronautics and Astronautics Cincinnati Student Chapter Vice President in 2010; participated in Students for the Exploration and Development of Space in 2009.	2009 - 2011
University of Cincinnati Astronomy Club Founding member in 2008, treasurer from 2008 to 2009, and secretary from 2009 to 2010.	2008 - 2010
Society of Physics Students Cincinnati Chapter Secretary from 2005 to 2006 and treasurer from 2006 to 2007.	2005 - 2008
Sigma Theta Epsilon Beta Gamma Chapter Held secretary, treasurer, chaplain, and publicity responsibilities.	2005 - 2008
Additional Academic Activities Participated in the Large Synoptic Survey Telescope seminar group, UC SPARKS, the University of Cincinnati Mountaineering Club, and University of Cincinnati physics departmental seminars from 2008 to 2010.	2006 - 2010
Athens Wargamers Member; served as treasurer during the 2002–2003 academic year.	Jan. 2001 - Jun. 2004

Creative and Visual Arts

Zygote Press Exhibiting Artist - Participated in printmaking exhibitions organized by the studio.	2011 - 2014
Senior B.F.A. Exhibition, Jennings House Gallery Exhibiting Artist - Presented senior printmaking work at Ohio University.	May 2004
Student Printmakers Exhibition and Print Exchange Organizer - Organized an exhibition and print exchange for student printmakers.	Feb. 2004
Lacuna Gallery, Logan, Ohio Exhibiting Artist - Displayed original artwork in a local gallery exhibition.	2004
Ohio University School of Art Commission Commissioned Artist - Created an original print for the School of Art president; the exact commission date and recipient name are not recorded.	2003 - 2004
Ohio University Visiting-Artist Print Projects Student Assistant - Assisted visiting artists Debra Fisher and Dan Murphy with two print projects.	2000 - 2004
Lakewood Hospital, Lakewood, Ohio Exhibiting Artist - Displayed artwork at the hospital.	1999

Selected Independent Projects

iphone-cam-lite Native Swift/SwiftUI macOS menu-bar app that selects and verifies iPhone Continuity Camera devices in Zoom and Microsoft Teams; includes permission-aware onboarding, diagnostics, persistent settings, automated tests, CI, and release packaging.	2025 - 2026
farm-invoice-parser Ruby, RubyLLM, OCR, and model APIs; converts handwritten farm invoices into structured transaction records.	2025
cf_d_snow_fence Python applied-modeling project combining measured wind data, Navier-Stokes pressure calculations, and XGBoost classification to identify potential snow-drift locations.	2025
healthcare-dbt-demo Runnable Data Vault 2.0 demonstration for healthcare claims data using dbt and DuckDB/PostgreSQL.	2025
tuva-postgres Python and PostgreSQL utilities for converting and validating Tuva seed data.	2025
claims_case_study Healthcare staffing-demand analysis using data-quality assessment, Parquet transformation, exploratory analysis, and ARIMA forecasting; delivered a written report and stakeholder presentation.	2023
HL7-ingestion Java and Docker healthcare-interoperability project converting HL7 messages into longitudinal patient records.	2023
quality_measures_engine Python project for healthcare quality-measure processing and evaluation workflows.	2023
DETL-design-challenge Health-data transformation and loading design demonstrating ETL automation and validation.	2023
zams_cluster_candidacy Python scientific-analysis pipeline using zero-age main-sequence and isochrone comparisons to evaluate stellar-cluster candidacy.	2019 - 2020
gamma_rays Python and Jupyter experiments in gamma-ray classification, energy reconstruction, and deep learning, with disseminated analysis outputs.	2019
Star-Chart JavaScript web application displaying the night sky for a selected location and date using stellar catalog data.	2015

Publications and Research Outputs

Reproducible Postgres Load of Tuva Seed Datasets	Aug. 2025
Reusable load and validation workflow for Tuva healthcare seed datasets.	
Surgical Scheduling Optimization and Procedure Duration Prediction	Dec. 2020
Optimization and prediction work improving on fixed-ratio methods for estimating total procedure time.	
Process and Requirements for CCD Data Integration	Oct. 2017
Processes and requirements for addressing challenges in CCD data curation.	
AI Paint Formula Prediction from Spectroscopic Measurements	May 2015
Naive Bayesian approach to color and effects matching for complex automotive paint formulas.	
Characterization of Surface Coatings Using RT Models for Color Matching	Feb. 2013
Approximation of a Chandrasekhar radiative-transfer model applied to atmospheric scattering and color matching.	
Massive Stellar Clusters in the Disk of the Milky Way Galaxy	Dec. 2010
Master's thesis presenting spectroscopic and photometric analysis of two open clusters in the Galactic plane.	
The B Stars of Westerlund 1	2009 - 2010
Near-infrared stellar-classification research manuscript prepared with M. M. Hanson; listed in contemporaneous resumes as in preparation.	
VizieR Online Data Catalog: JHK Photometry in Cluster [BDS2003] 107	May 2008
Near-infrared photometric catalog associated with observations of the Galactic cluster.	
Near-Infrared Spectroscopy of Candidate Members of Galactic Cluster [BDS2003] 107	Feb. 2008
Classification spectra for nine candidate cluster members, published in Publications of the Astronomical Society of the Pacific, 120(864), 150–159.	
A Novel Method for Low-Toxic Photo-Etch Printing	Fall/Winter 2003
Published by the Mid America Print Council (MAPC), Volume 11, Number 2; presented undergraduate research applying aquatint etching techniques to photo-etch procedures.	

Speaking and Knowledge Sharing

AI Hackathon: Surgery Scheduling Optimization	Dec. 2020
Presented an approach using historical utilization and current wait-list information to predict surgery duration.	
REACH Initiative: IBM Watson Care Insights	2019
Presented the use of patient records and real-time HL7 feeds to integrate insights into clinical workflows.	
Content and Data Analytics Learning Exchange	2018
Introduced machine-learning implementation using real-world health-data analytics.	
Innovation Research Interchange Fall Networks Conference	Sep. 2018
Presented natural-language processing for extracting insight from unstructured data lakes.	
Cross-Team Training Series	2016
Led a seven-part series on Hadoop ingestion and data-lake concepts.	
M.S. Physics Thesis Defense, University of Cincinnati	Jun. 2010
Defended a thesis presenting spectroscopic and photometric analysis of two open clusters in the Galactic plane.	
2MASS Research Poster Session	2008
Presented a poster on astronomical research using Two Micron All Sky Survey (2MASS) data.	

Professional Recognition

Special Recognition, Health Data Movers - client delivery	2025
Arkos Health BI Star - Salesforce Trailhead Expeditioner	2023
Manager Recognition for Transition Leadership, Merative - systems migration	2022
Gold Champion, IBM - 1,477 classroom hours and 43 certificates	2018 - 2022
Lab Services Award, IBM - cross-team delivery coordination	2018
Eminence and Excellence Award, IBM - data-curation tool development	2017
Color Lab Automation Award, PPG Industries - expanded reporting	2012
Mary J. Hanna Research Fellowship, University of Cincinnati	2010
Academic Scholarships and Honors	2000 - 2011
University Graduate Scholarships for physics study from 2008 to 2010 and aerospace-engineering study in 2011; University of Cincinnati Dean's List from 2005 to 2007; Ohio University Provost's Undergraduate Research Grant in 2003 and Dean's List from 2002 to 2004; Ohio Proficiency Scholarship in 2000.	

Education

Graduate study, Aerospace Engineering, University of Cincinnati, 2011 - Computational Fluid Dynamics; degree not completed	
M.S. Physics, Astrophysics concentration, University of Cincinnati, Dec. 2010 - thesis option; GPA 3.3/4.0	
B.F.A. Printmaking, Ohio University, Aug. 2004 - major GPA 3.6/4.0	
Additional coursework: Windows Forms; quantitative finance; information risk; computer networks; computational data analysis; AI planning; astrobiology; critical thinking	

Professional Development and Credentials

Fundamentals of Visualization with Tableau - University of California, Davis

Jun. 2023

Credential ID M3VVMS4BTXWU.

IBM Data Science Professional Certificate

2022

Certificate of Specialization, Data Science - Johns Hopkins University

2015

IBM Digital Credentials and Coursework

Nov. 2017 - Feb. 2022

Completed 43 distinct IBM credentials spanning data science, machine learning, big data, cloud, product management, design thinking, security, blockchain, and quantum computing.

- **Nov. 2017:** IBM Agile Explorer.
- **Feb. 2018:** IBM Blockchain Essentials.
- **May 2018:** IBM Cloud Private Infrastructure and Architecture; Hadoop Data Access - Level 1.
- **Jun. 2018:** Product Manager - Define, Build & Deliver; Manage the Business; Product Manager - Scaling the Business.
- **Aug. 2018:** Cognitive Practitioner.
- **Feb. 2020:** IBM Quantum Conversations.
- **Sep. 2020:** AgileTA Explorer; Think Like a Hacker.
- **Oct. 2020:** Watson Visual Recognition; Enterprise Design Thinking Practitioner; IBM Cloud Private - Foundation Technology; Watson and Cloud Foundations; Enterprise Design Thinking Co-Creator.
- **Jan. 2021:** Cloud Service Management and Operations Explorer.
- **Jun. 2021:** Data Science Foundations - Level 1.
- **Jan. 2022:** Data Science Methodologies; Python for Data Science; Hadoop Programming - Level 1; Hadoop Foundations - Level 1; Cloud Core; Data Analysis Using Python; Simplifying Data Pipelines with Apache Kafka; Deep Learning Essentials; Spark - Level 1; Data Science with Scala; Applied Data Science with Python - Level 2; Data Visualization Using Python; Machine Learning with Python - Level 1; Data Science Tools; IBM Blockchain Essentials V2; Introduction to Machine Learning with Sound; Big Data Foundations - Level 2; Big Data Foundations - Level 1; Machine Learning with R - Level 1; Data Visualization with R; Deep Learning using TensorFlow; Deep Learning; Accelerated Deep Learning with GPU; Data Science Foundations - Level 2 (V2).
- **Feb. 2022:** IBM AI Associate Data Scientist.

Salesforce Trailhead Coursework Completed at Arkos Health

2023 - 2025

Reached Trailblazer Mountaineer rank with 18,725 points and 40 badges.

- **Tableau and data literacy:** Tableau Prep Builder Basics; Tableau Blueprint Basics; Tableau Visualization Formatting on the Web; Tableau AI and Analytics: Quick Look; Data Sharing and Collaboration in Tableau Cloud; Data Visualization in Tableau Cloud; Tableau Cloud Basics; Data Presentation and Publication in Tableau Desktop; Data Filtering and Sorting in Tableau Desktop; Data Analysis in Tableau Desktop; Data Connection in Tableau Desktop; Tableau Data Management: Quick Look; Variation for Data Comparisons; Data Distributions; Aggregation and Granularity; Variables and Field Types; Well-Structured Data; Data Literacy Basics; The Tableau Desktop Workspace; The Tableau Workflow.
- **Salesforce, analytics, AI, and security:** User Management; Happy Birthday Trailhead 2023; Salesforce CRM; User Engagement; Lightning Experience Customization; Data Modeling; Agentforce 360 Platform Basics; Formulas and Validations; Correlation and Regression; Data Analytics Fundamentals; Natural Language Processing Basics; Artificial Intelligence Fundamentals; Generative AI Basics; Responsible Creation of Artificial Intelligence; Superbadge Program Security: Quick Look; Lightning Experience Basics; Data Management; CRM Analytics: Quick Look; Data Security; Reports & Dashboards for Lightning Experience.